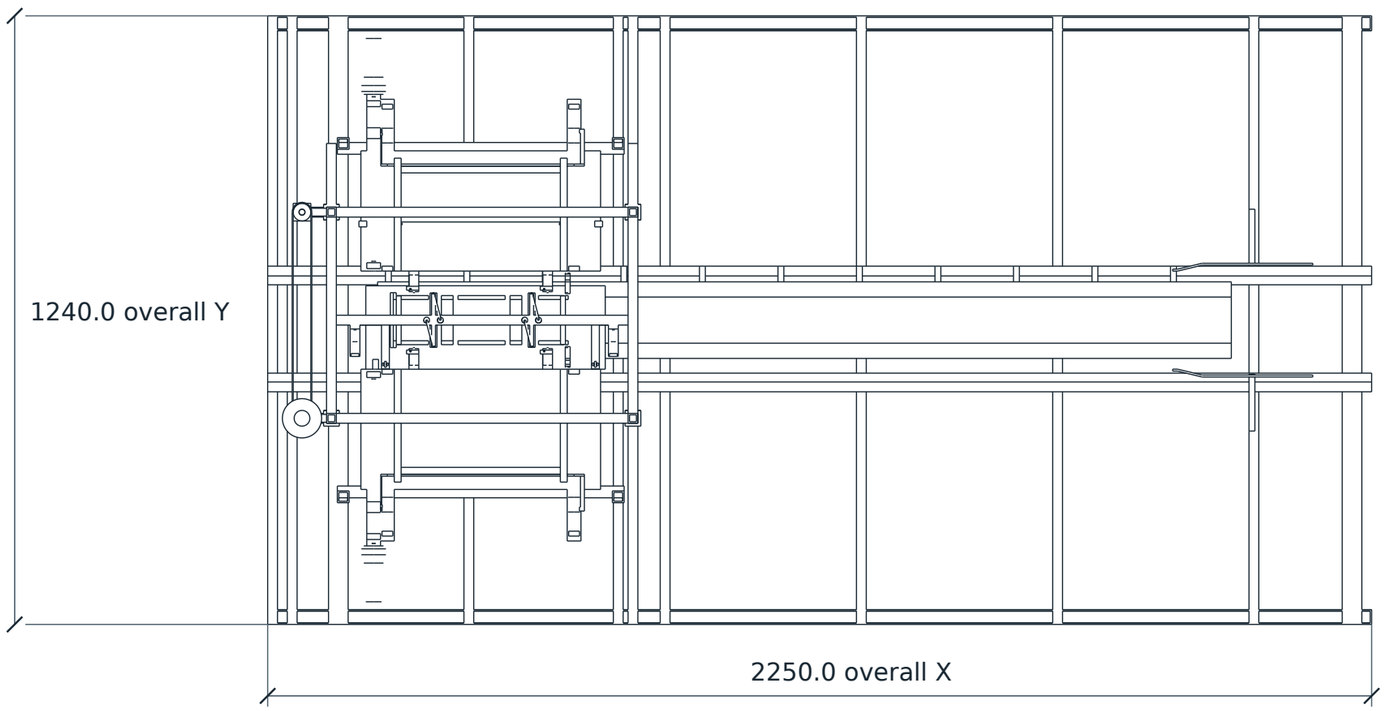
VOLLEY-G5-ACR1

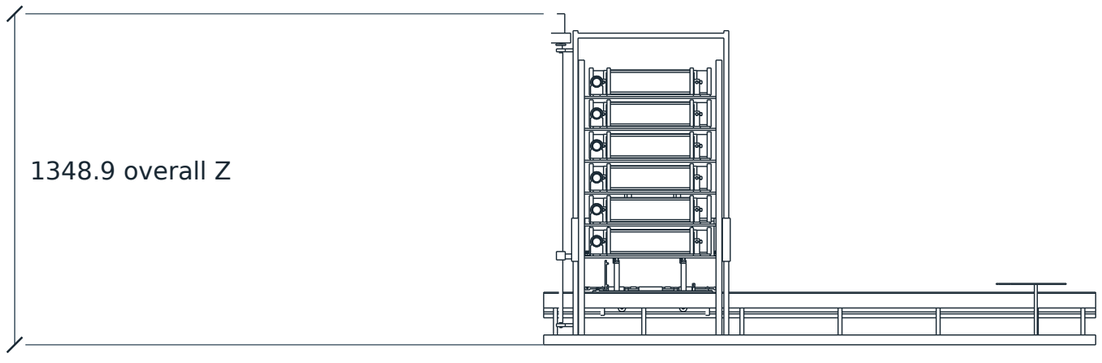
General arrangement — complete home-state envelope

OUTLINE / PRELIMINARY ENGINEERING — dimensions in mm; no general manufacturing tolerance assigned

PLAN: +Z projection | scale 1:12.5



SIDE: X-Z projection | scale 1:25



Home envelope from native BRep:
X 0.0 to 2250.0
Y -620.0 to 620.0
Z -190.0 to 1158.9

12 × 4 kg 3U envelope surrogates
2 fixed banks × 6 levels

Projected home: 387 connected parts
Powered travel: 1300
Carrier aft X: 200 → 1500
Arrest allocation: 210
Rail X: 0 → 2250

Reference mounting plane Z = -150
Generic host ladder included.
No provider hole-pattern approval.

D01 | Revision A | 25 October 2026 | A3 landscape

Source: frozen ACR1 native geometry and paired BOM/verification. Do not scale; stated dimensions govern.

Academic engineering outline

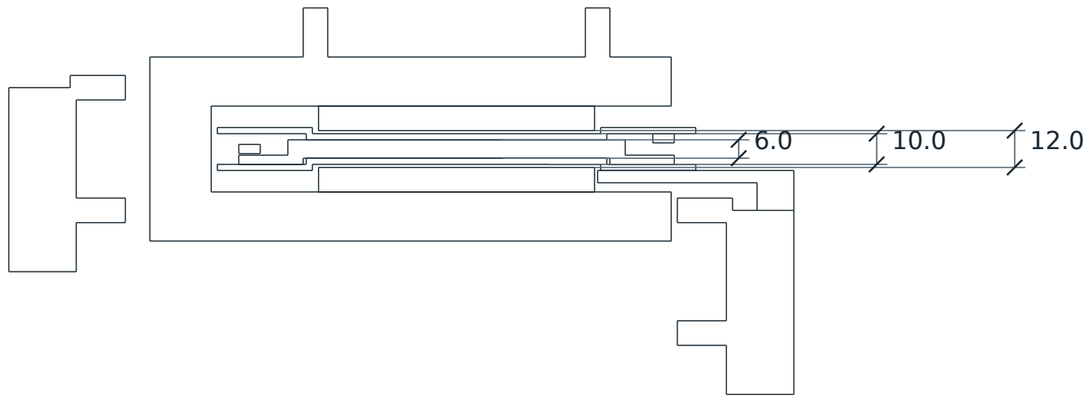
NOT A MANUFACTURING RELEASE

VOLLEY-G5-ACR1

Motor section and launch datums

OUTLINE / PRELIMINARY ENGINEERING — dimensions in mm; no general manufacturing tolerance assigned

SECTION A-A at X = 400 | scale 1:2



Nominal transverse stack (Z):
Upper magnet inner face +6
Upper GFRP outer face +5
Upper copper face +3
Lower copper face -3
Lower GFRP outer face -5
Lower magnet inner face -6

Each magnet/package face gap: 1
Copper active thickness: 6
GFRP: 2 each active face
Active bar: 7 × 6 = 42 mm²
Return minimum: 7 × 3 = 21 mm²

These are CAD nominal dimensions.
Airgap budget is on sheet D05.
No blanket fit/flatness grade set.

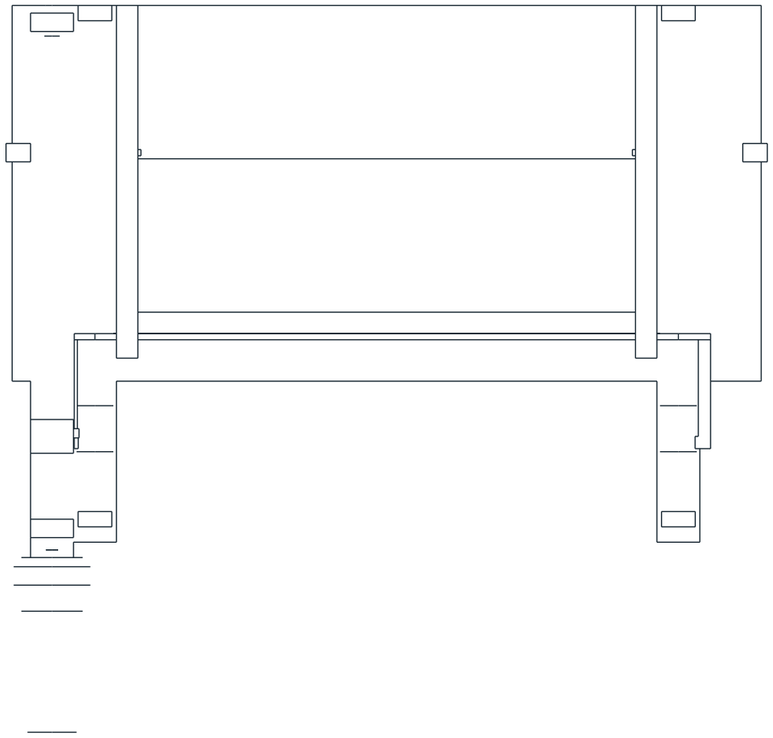
Active magnet X relative to carrier aft: 76 to 412 (336 active; 340 mounting envelope).
Home active X: 276 to 612. Release active X: 1576 to 1912.
216 stator belts at 8 pitch: X 230 to 1958; end allowance 46 at home/release.
Payload floor Z = 46. Carrier top plate Z = 14 to 30; bottom plate Z = -30 to -14.
Native rails: -Y bounds Y -146 to -108, Z -40 to 24; +Y bounds Y 46 to 110, Z -80 to -7.

VOLLEY-G5-ACR1

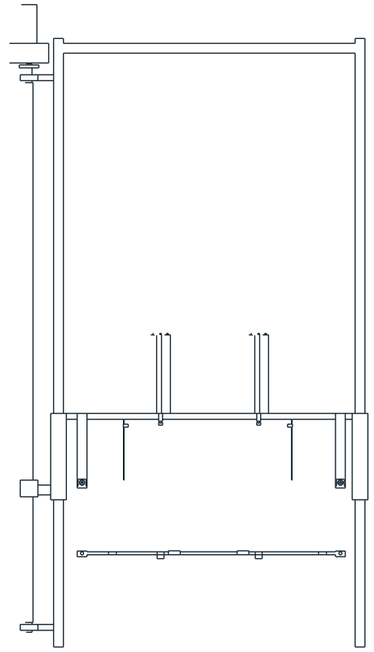
Cassette feed and lift interfaces

OUTLINE / PRELIMINARY ENGINEERING — dimensions in mm; no general manufacturing tolerance assigned

South bank level 0: PLAN | scale 1:4



Lift/pallet: SIDE | scale 1:12.5



Cell floors Z: 180, 310, 440, 570, 700, 830 (pitch 130).

Payload 340.5 × 100 × 100; channel side/top clearance 2.

Cell centers Y = ±250; feed stroke 250 in 40 s (study).

Guides Ø10; span 330. Screw Ø10, lead 2, study root Ø7.4.

South guide Y = -435 to -105; front journal -120 to -100.

Lift docking floor Z = 46; highest floor 830 → travel 784.

Twin screw nominal Ø16, lead 4; study root Ø11.

Fixed screw axes Y = -200 and +220.

Reference gearbox ratio 19.8; maximum lift time 120 s.

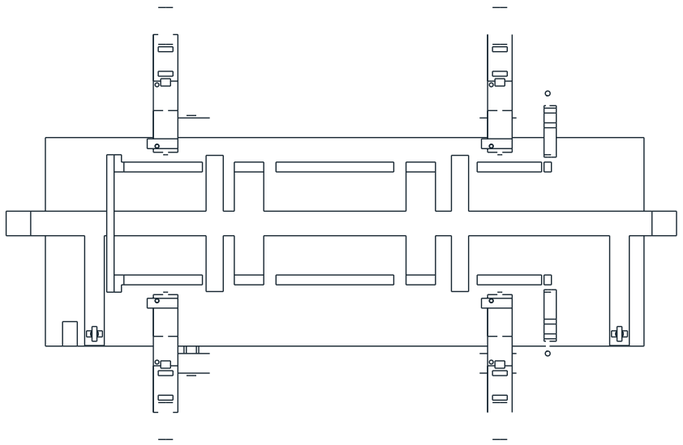
Reference actuator envelopes do not define supplier flanges.

VOLLEY-G5-ACR1

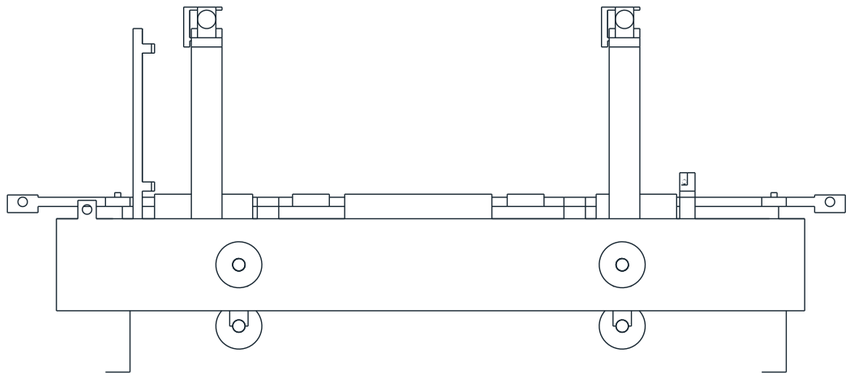
Sled, captive pallet and custody interfaces

OUTLINE / PRELIMINARY ENGINEERING — dimensions in mm; no general manufacturing tolerance assigned

Retained sled/pallet: PLAN | scale 1:5



Retained sled/pallet: SIDE | scale 1:4



Carrier home aft X = 200.

Carrier nominal length 488.

Full home assembly X:

168 to 714.5 with pallet hardware.

Pallet lock relative X: 40 / 468

Lock axis Y = -90

Carrier shaft bore Ø6.2

Stem Ø6; entry round bore Ø7

Entry key slot 13 × 5

Bayonet stroke 12; rotation 90°

Lift grip X: 178 / 704.5

Grip pin Ø6; bore Ø6.2

Grip withdrawal stroke Y: 26

Pallet feet Z = 30 to 38

Payload rail contact Z = 46

Flight clearance nominal 0.5

Custody: land → close capture caps/lockpins → insert/rotate both bayonets → prove → open keepers → withdraw grips → raise clear.

VOLLEY-G5-ACR1

Datum, tolerance-budget and assembly inspection notes

OUTLINE / PRELIMINARY ENGINEERING — dimensions in mm; no general manufacturing tolerance assigned

PROJECT REFERENCE DATUMS (preliminary; no supplier datum scheme assigned)

A: generic mounting plane Z = -150. B: launch direction +X. C: launcher median plane Y = 0.

Dimensions are basic CAD nominal values. A general ± tolerance or GD&T release is not assigned.

MAGNET/PACKAGE INDIVIDUAL-FACE STACK — declared allocation, not a measured fit

Nominal 1.000 − FE paired closure 0.371778/2 − guide centring 0.200 − flatness 0.150

− assembly 0.150 − differential thermal 0.050 = 0.264111 mm residual per face.

FE closure uses the declared 130% load screen; load/material/mesh limits remain in the source report.

CRADLE ENDWALL STACK — each nominal axial gap 0.500

Machining allocation 0.100 + docking allocation 0.100; differential CTE assumption 12 µm/(m·K).

340.5 mm reference length: residual 0.300000 / 0.218280 / 0.136560 / 0.054840 mm

at differential temperature 0 / 20 / 40 / 60 K. Elastic contacts need later correlation.

ASSEMBLY AND INSPECTION SEQUENCE

1 Identify each part against BOM_HOME.csv and frozen native labels; verify material class and mass assignment.

2 Establish project mounting and rail datums; inspect rail straightness/parallelism within the allocated stack.

3 Assemble insulated stator and sled; inspect both face gaps over home-to-release travel before energizing.

4 Verify cassette guide/channel clearance, screw support alignment and each of twelve prescribed transfer routes.

5 Land pallet; inspect caps, lockpins, both bayonet/grip overlaps and capture-before-release order.

6 Inspect cam opening/running clearances and full cradle withdrawal; execute powerless/manual custody checks.

7 Record as-built dimensions, exceptions and inspection instrument calibration before a hardware test release.

Acceptance checklist: INSPECTION_PLAN.csv. No inspection, material certification, torque or physical test is recorded here.

Mass properties, material classes and fabrication boundary

OUTLINE / PRELIMINARY ENGINEERING — dimensions in mm; no general manufacturing tolerance assigned

HOME native assembly: 387 parts; loaded 197.556720 kg = dry 149.556720 + payload 48.000000.

Represented dry CG (mm): X 628.361876; Y -4.213144; Z 267.524868.

Loaded native CG (mm): X 581.198863; Y -3.189484; Z 337.372183.

Retained sled with pallet: 13.127806 kg. Mission device: 156.556720 kg; loaded device: 204.556720 kg.

Extra 7.000 kg power/control allowance has NO assigned position and is excluded from CAD CG/inertia.

Represented dry inertia about its own CG [kg·m²], full tensor, source sign convention:

[29.675715228 0.060095224 12.702423066]
[0.060095224 62.806994278 0.035579787]
[12.702423066 0.035579787 54.885372840]

Materials are assigned density classes; no alloy temper, coating, bond specification or supplier certification is assigned.

Aluminium 2700; steel 7850; copper 8960; NdFeB 7500; polymer 1410; insulator 1800 kg/m³.

Catalog/payload envelope mass distributions remain approximate; see exact per-group source boundaries.

BOM_HOME.csv contains all native part labels, assigned masses, CG coordinates, volumes and source notes.

MASS_PROPERTIES.json retains all eight native-state accepted group CG/inertia records.

PRELIMINARY FABRICATION ROUTE

Frame/pallet: stock machining and joint/inspection planning remain to be detailed from the owned solid model.

Copper phases: preserve connected topology, minimum return section and insulation gaps; joining/insulation process unselected.

Guides, screws, bearings, brakes and actuators: source envelope/reference only where catalog interfaces are unselected.

No fastener torque, interference fit, vacuum material compatibility or supplier acceptance is invented by these sheets.